



Project

MIRYOR

An AI Powered Memory and Emotional Analytics System

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Outline

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Problem Statement

We Forget Things Fast: :

Our brains naturally forget deep thoughts and life insights within a few days.

Standard Apps are Passive::

Apps like Notion, Google Keep, or WhatsApp just hold text like an empty box. They cannot connect your past notes to your present life.

AI Apps Have Paywalls:

Existing AI tools like Rosebud AI or Mindsera are locked behind expensive monthly subscriptions.

The MIRROR Solution:

A smart, private AI journal that connects your history to your present. When you chat with it, MIRROR instantly retrieves relevant past entries, resurfacing your old life lessons right when you need them, completely free to use.



Technology Stack

The Frontend:

Built with Ionic and Angular, allowing the exact same code to run smoothly on both web browsers and Android phones.

The Backend:

Written in Java Spring Boot and divided into microservices to keep the system fast, scalable, and organized.

The Database:

Powered by PostgreSQL with the pgvector extension, which allows us to store standard text and AI embeddings together.

The AI Connection:

Uses the Gemini API to understand the deep meaning of your text, and the Groq API to generate lightning-fast conversational answers.



Core AI Engine

Saving Thoughts (Embeddings):

You type or speak a note into the app. The Gemini API acts as a translator, converting your words into mathematical numbers (vector embeddings) to capture the true meaning and emotion, not just the text.

Smart Memory Match (Semantic Search):

Our database acts like a smart librarian. Instead of matching keywords, it uses semantic search to compare those numbers and instantly find old notes where you faced similar situations.

Personal Advice (RAG):

We use a system called Retrieval Augmented Generation. The Groq API acts like a wise friend, it reads your new note alongside those past memories, then instantly replies with advice based on your own life lessons.



Live Demonstration



Conclusion & Future Scope

Conclusion:

Problem Solved: MIRROR successfully transforms passive note-taking into an active, conversational AI experience.

Accessible Innovation: Proved that premium, personalized AI features can be built entirely for free using modern developer tools.

Simplicity & Privacy: Delivered a clean, focused alternative to bloated commercial dashboards.

Future Scope:

Multimedia Memories: Allow users to upload photos, using AI vision to understand and connect visual moments to their journal entries.

Proactive AI Check-ins: Evolve the AI to act like a true friend. If you write about being stressed, the AI will proactively check on you a few days later to see how you are doing.

Personal Life Coach: The AI will automatically identify your goals from your daily notes and act as a smart accountability partner to help you achieve them.



Thank You

Repository: <https://github.com/prabhu-tejaa/MIRROR>
Web Interface: <https://projectmirror.tech/>
Android Build: <https://github.com/prabhu-tejaa/MIRROR/releases>

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